

Lecture 1 — Atherosclerosis and Peripheral Vascular Disease

Companion Figures

Prof. Joseph Boyle · Imperial College London

1-BRS-CVR-6 · 1-BRS-CVR-8



Figure L1.1 — Vortex lift above Concorde wing; analogy for eddy flow at arterial branches and bends.

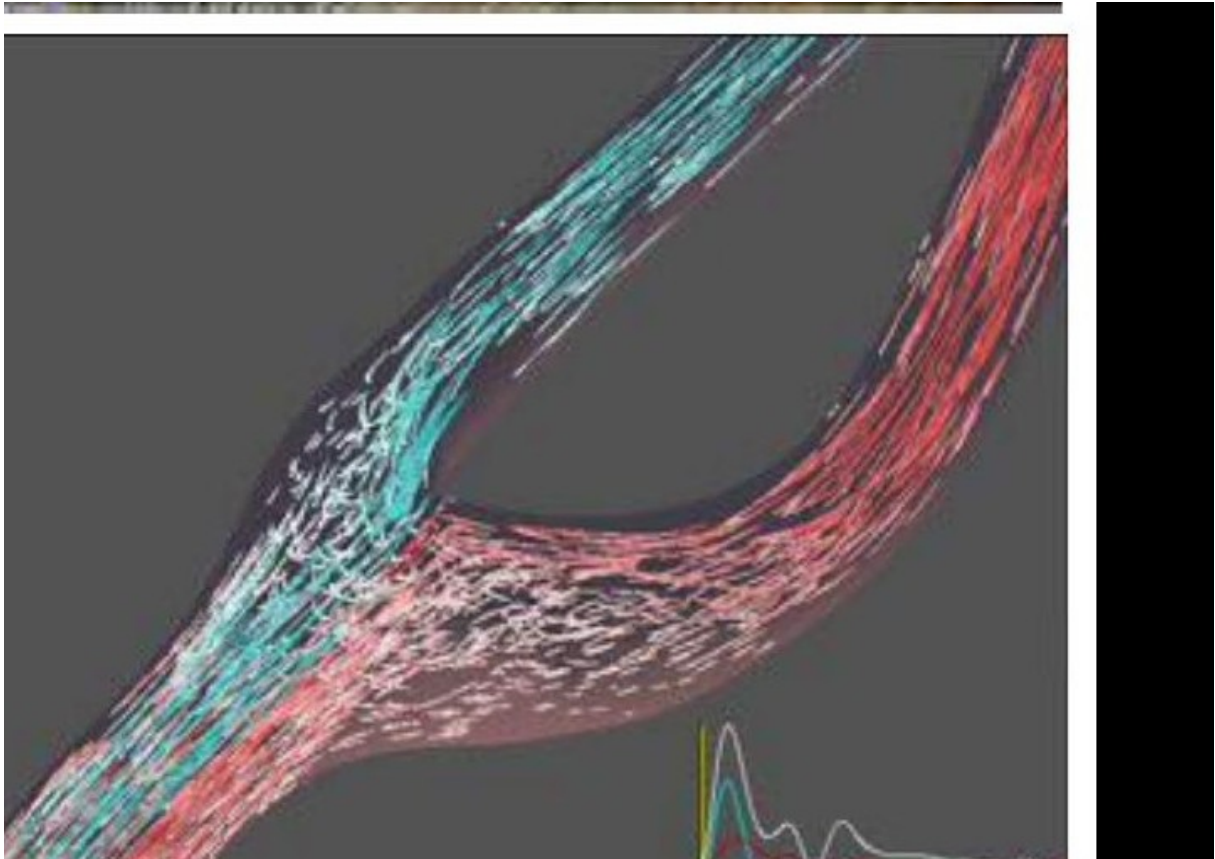


Figure L1.2 — Computed streamlines in carotid bifurcation showing eddies at the branch point.

ity is atherosclerosis local?

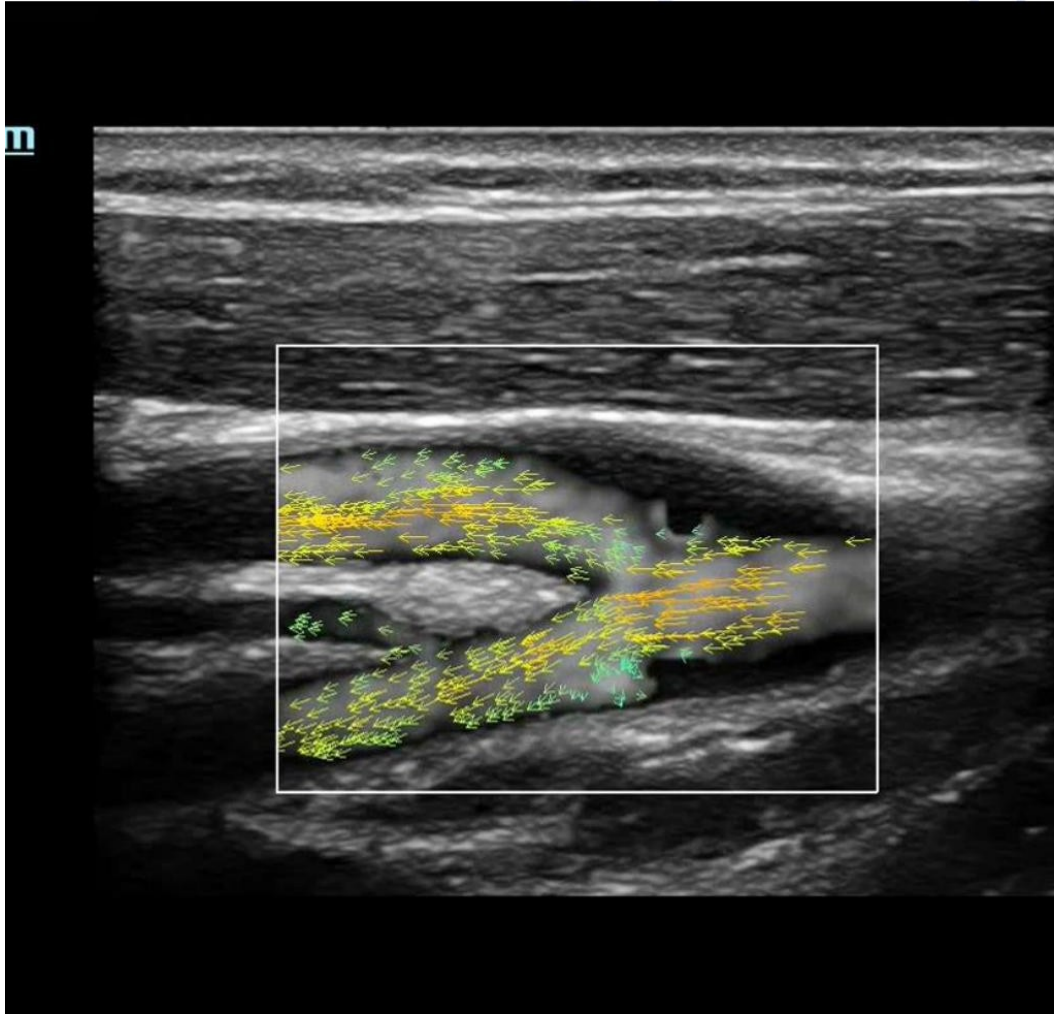


Figure L1.3 — Colour Doppler ultrasound of a live carotid bifurcation showing velocity vectors.

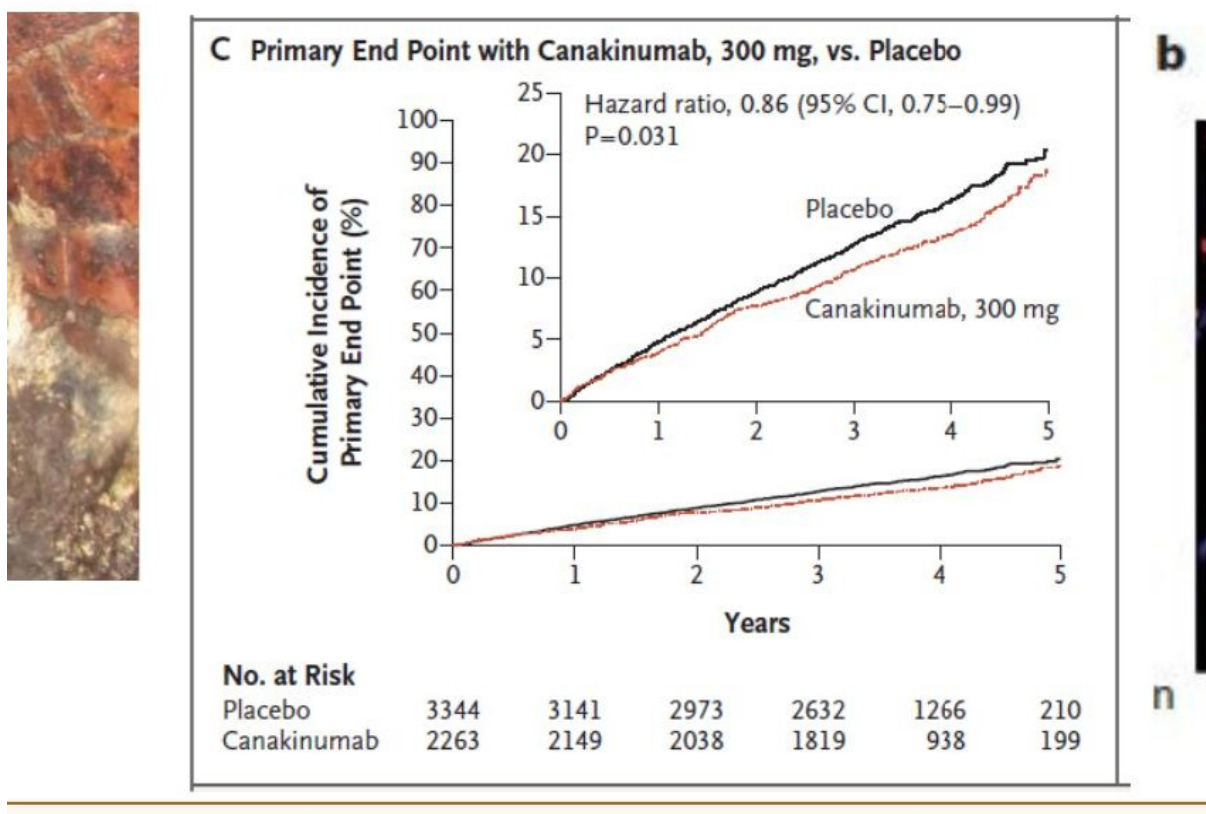
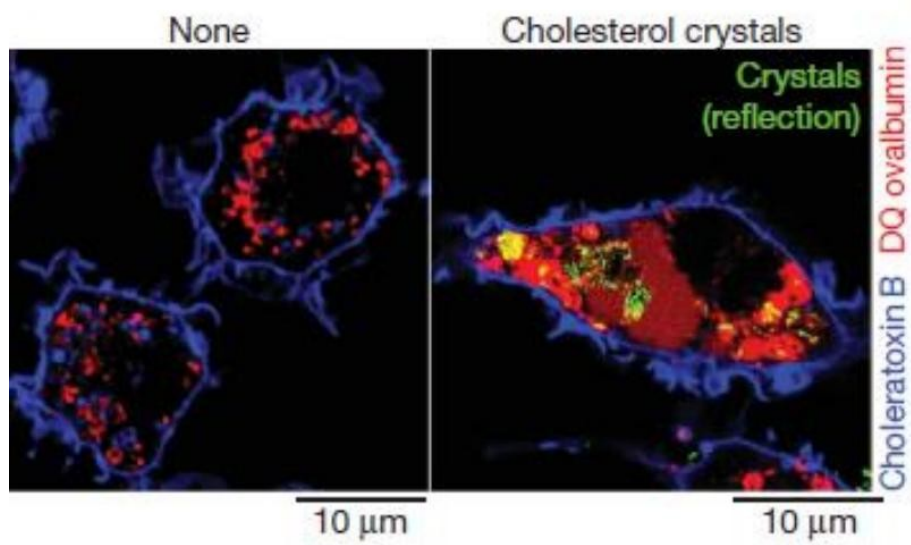
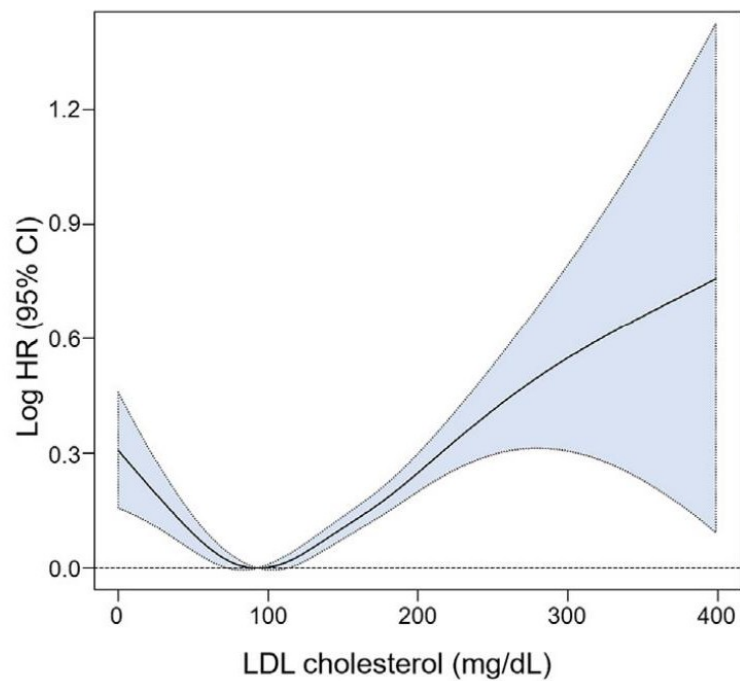
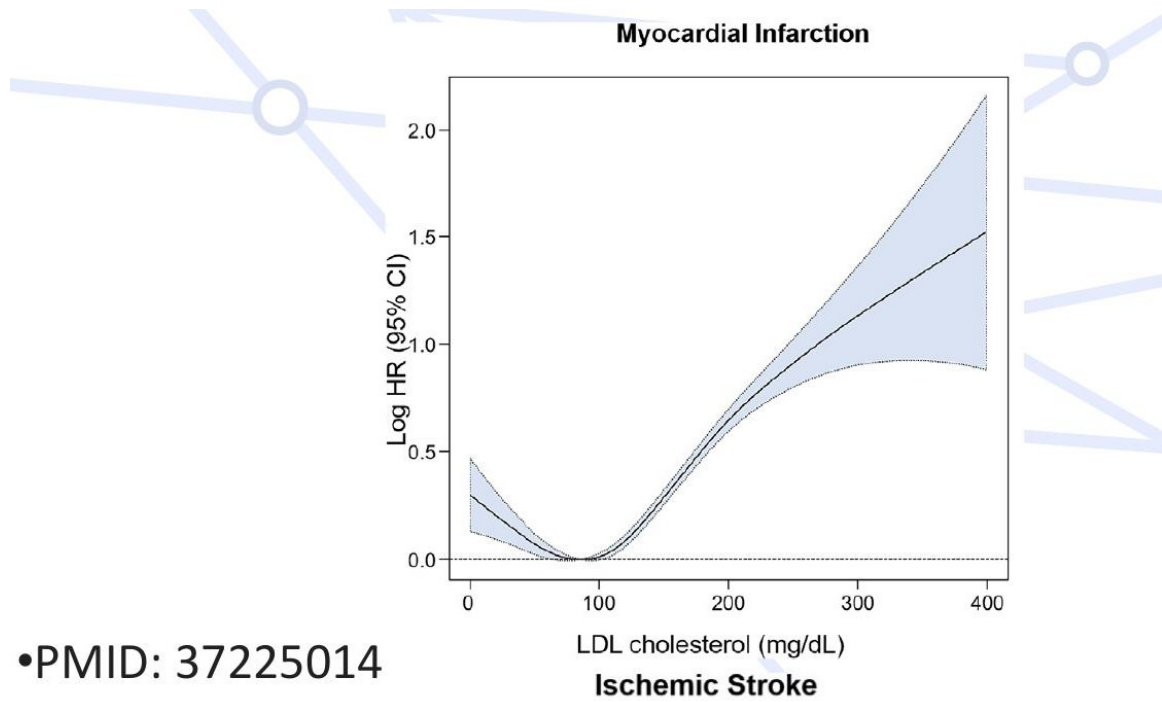


Figure L1.4 — CANTOS trial: cumulative incidence of primary end-point (MACE), canakinumab 300 mg vs placebo (HR 0.86, $p = 0.031$).



Pathophysiology of cardiovascular disorders

Figure L1.5 — Macrophages without (left) and with (right) cholesterol crystals, showing intracellular crystal reflection.



athophysiology of cardiovascular disorders.

Figure L1.6 — J-curve relationship between LDL cholesterol and risk of myocardial infarction (top) and ischaemic stroke (bottom).

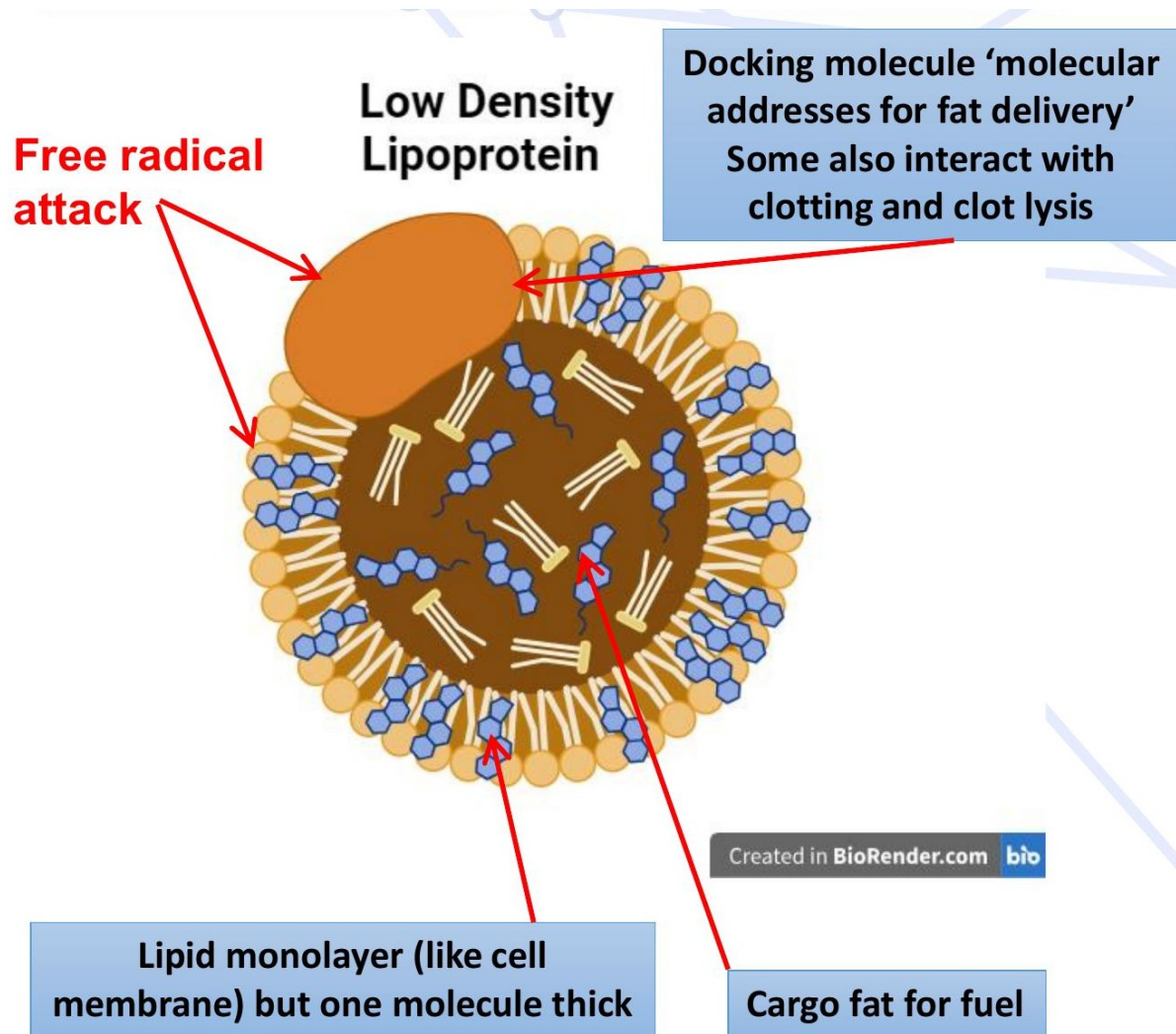


Figure L1.7 — LDL particle structure: lipid monolayer (one molecule thick), apolipoprotein docking molecule, and cargo lipid core.



Figure L1.8 — Tendon and cutaneous xanthomas in a child with homozygous familial hypercholesterolaemia.

Chapman (11)

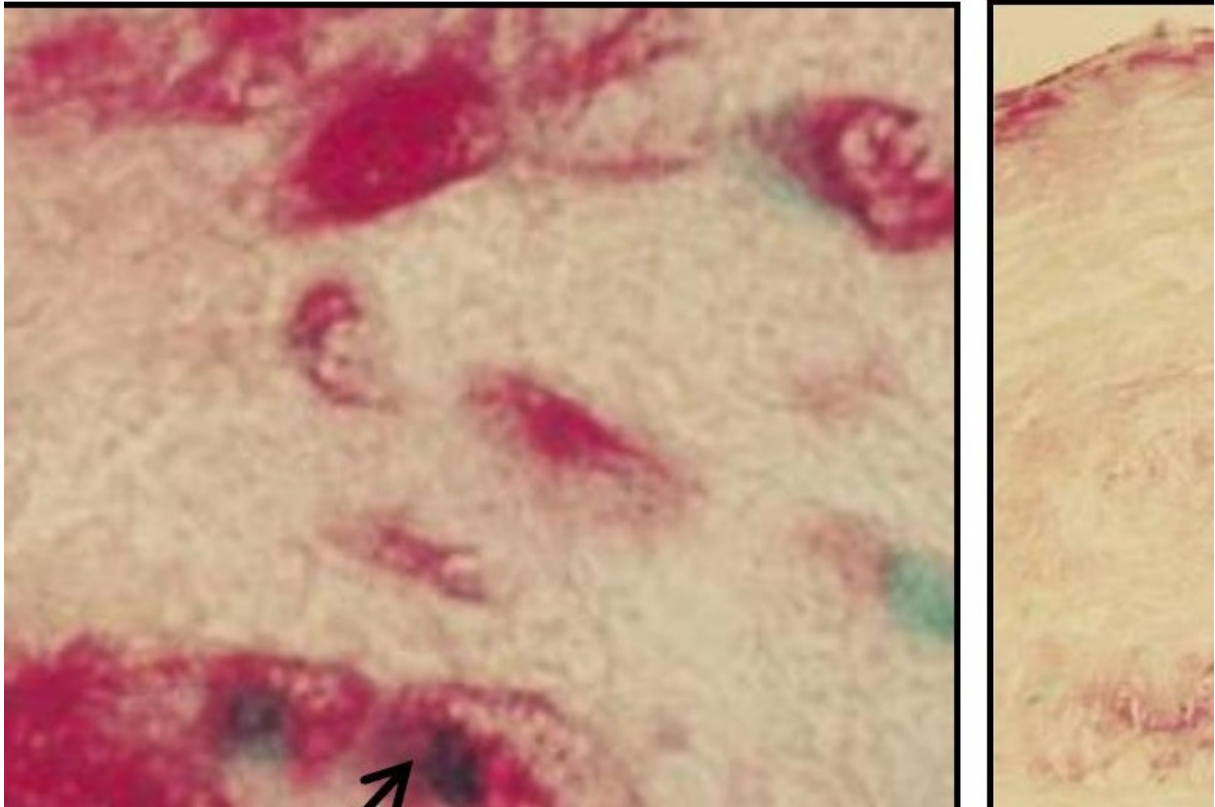


Figure L1.9 — Foam cells in a familial hypercholesterolaemia plaque (lipid stain).

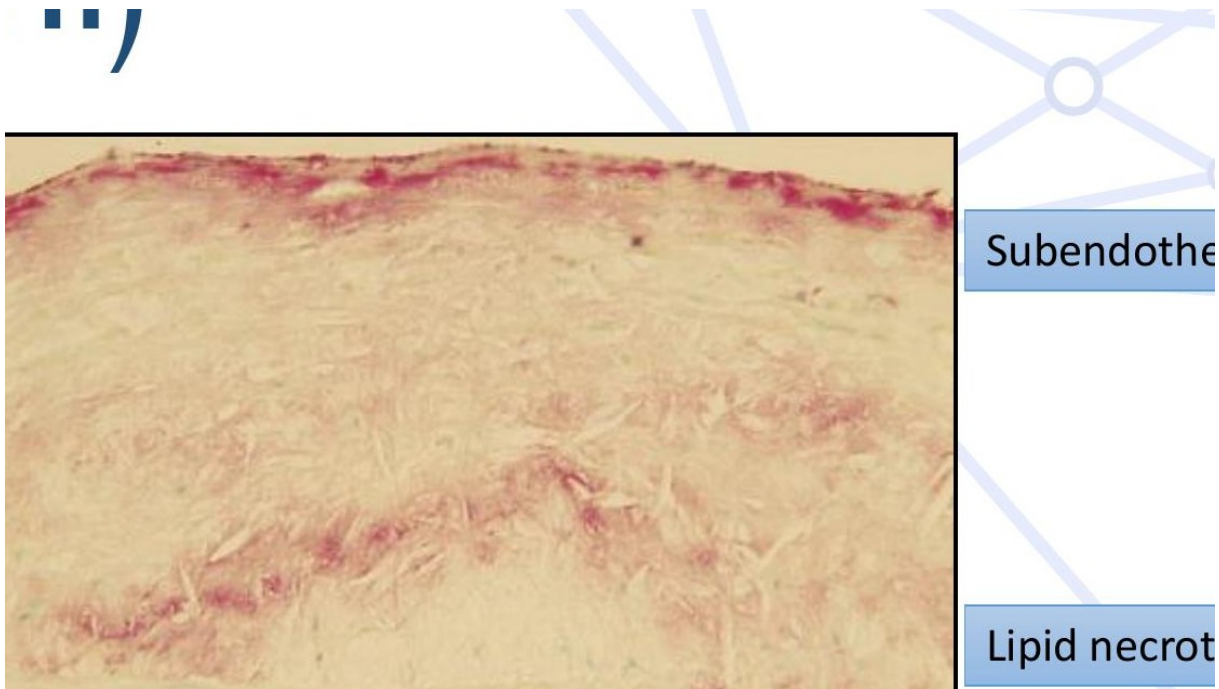


Figure L1.10 — Familial hypercholesterolaemia plaque: subendothelial lipid accumulation above a lipid necrotic core.

modified LDLs to become enlarged foam cells.

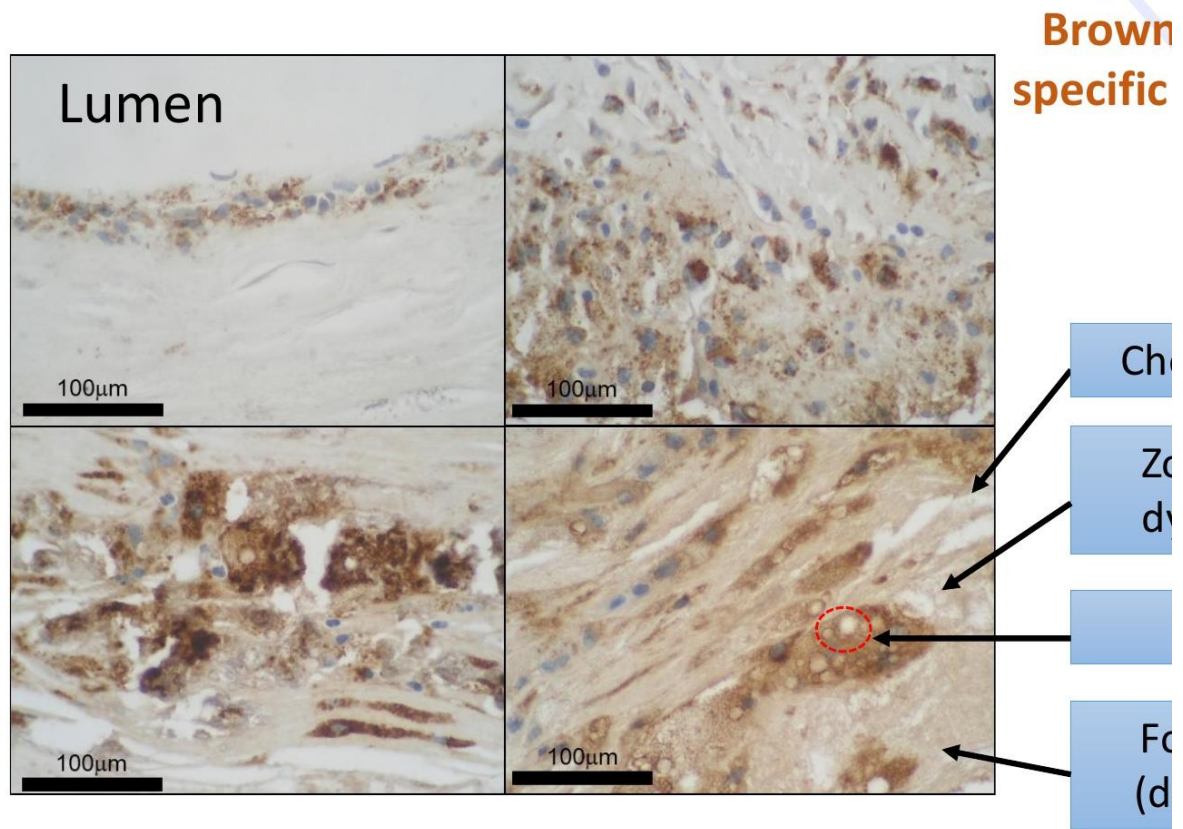


Figure L1.11 — Progressive plaque histology stained for CD68 (macrophages, brown): early lumen lesion progressing to foam cells, cholesterol crystals, sick/dying zone, and foam cell debris. All panels at the same magnification (100 µm scale bars).

(=MMPs)

us enzymes.
proteolysis.

sed on Zn.



C

Figure L1.12 — Coronary plaque: macrophage distribution (left) versus collagen distribution (right). Inverse co-localisation demonstrates MMP-driven collagen degradation (Van der Wal 1999).

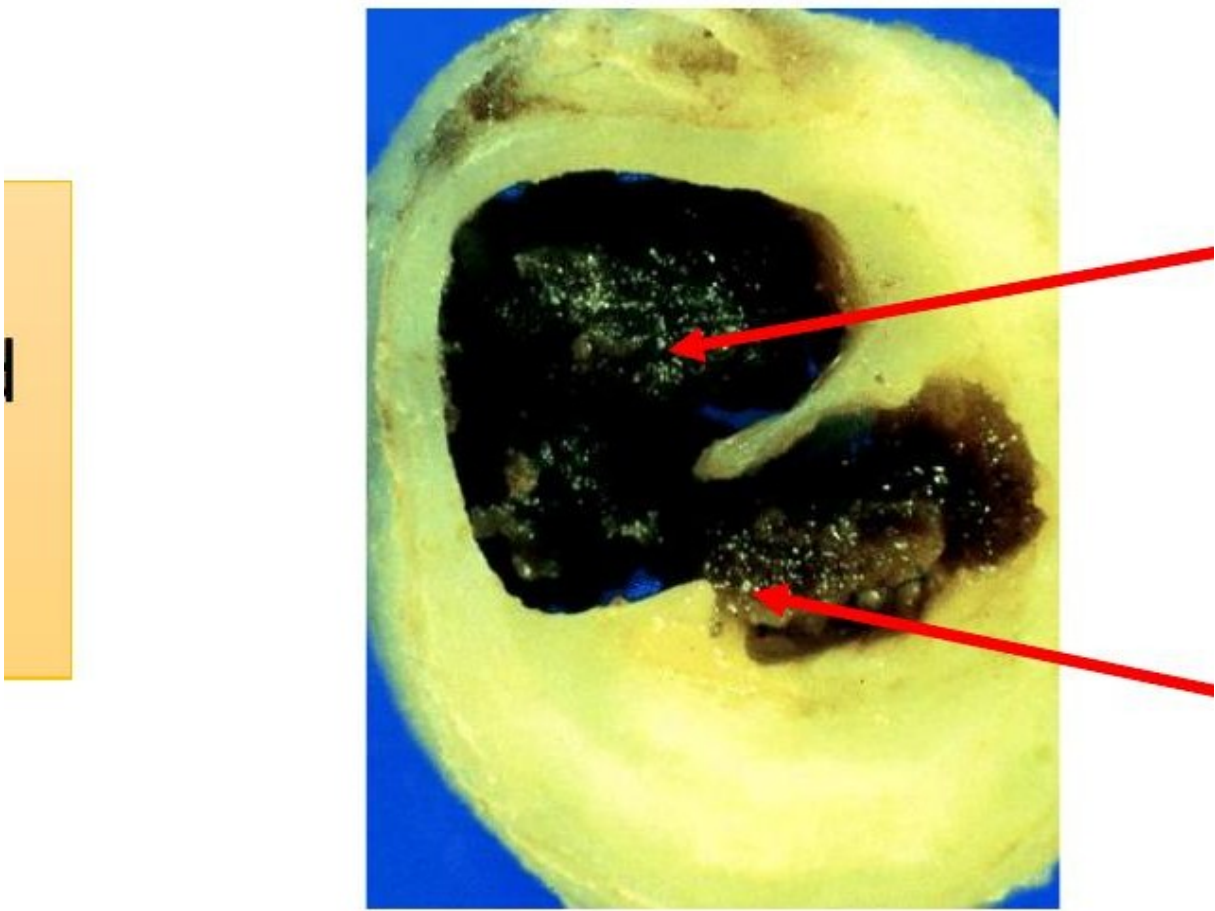
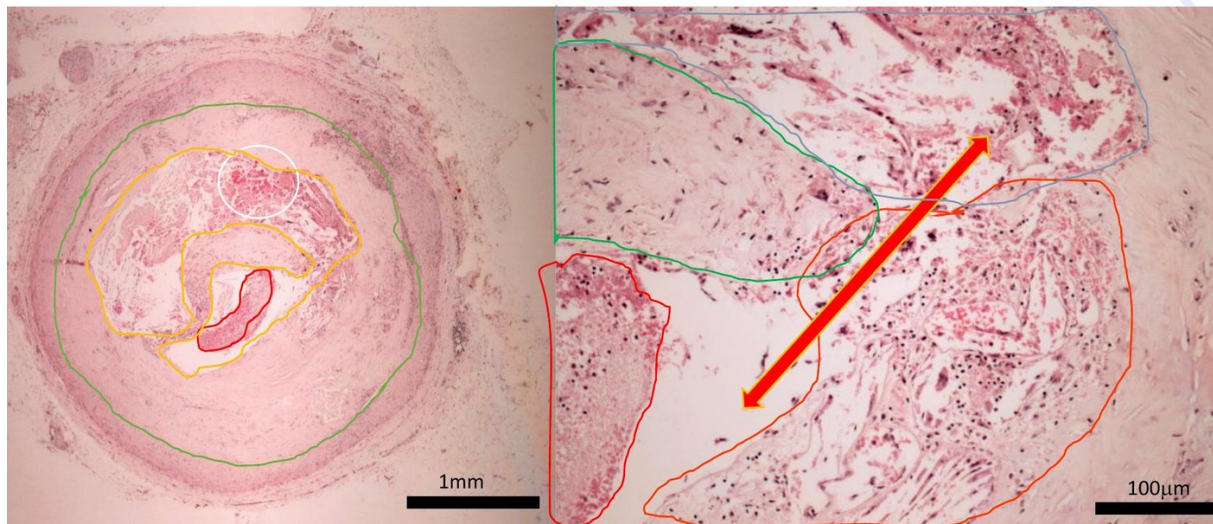


Figure L1.13 — Gross cross-section of a ruptured atherosclerotic plaque with an occlusive thrombus and visible site of rupture.



General staining

Thrombus

Inflammatory
macrophage infiltrate

Path 1007

Figure L1.14 — Ruptured plaque H&E; (J. Pathol. 1997): lipid necrotic core, thin fibrous cap, thrombus filling the lumen, and a dense inflammatory macrophage infiltrate at the rupture site.

Thrombus occluding left anterior
descending coronary artery

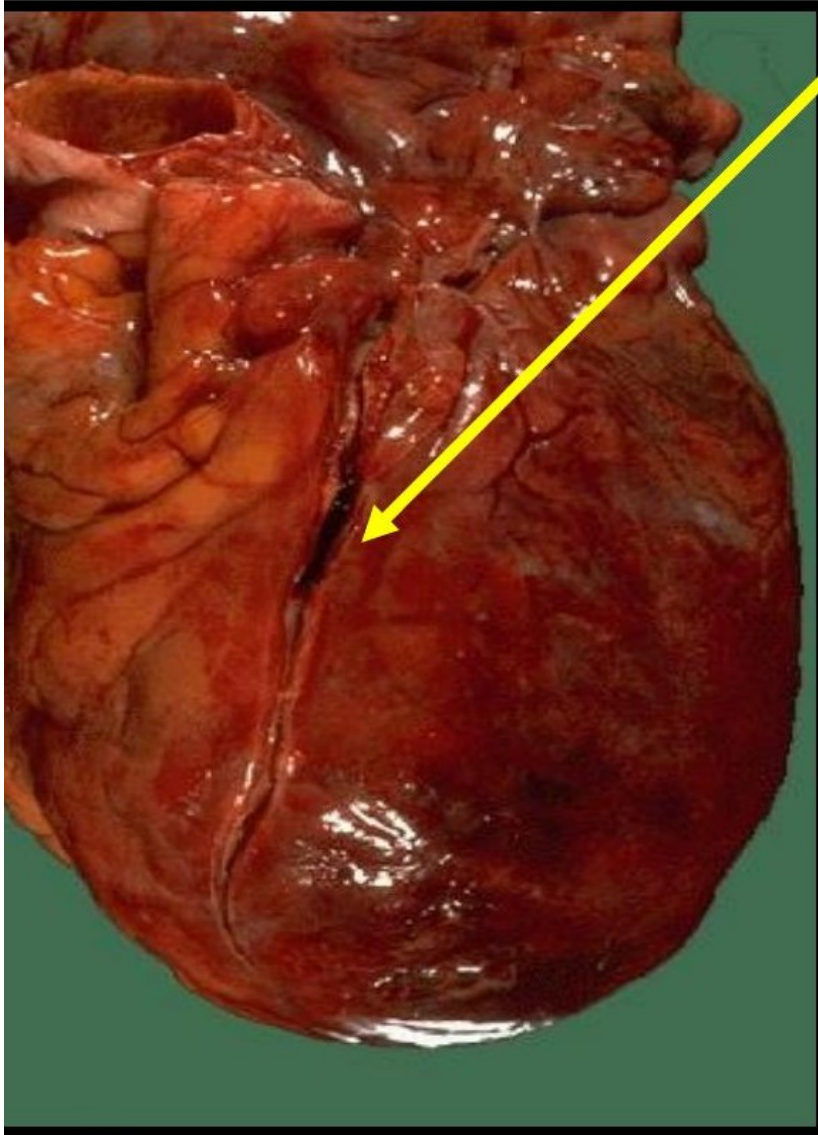


Figure L1.15 — Thrombus occluding the left anterior descending (LAD) coronary artery at autopsy.

Anteroseptal myocardial infarction

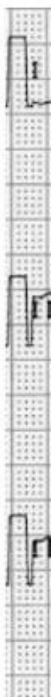
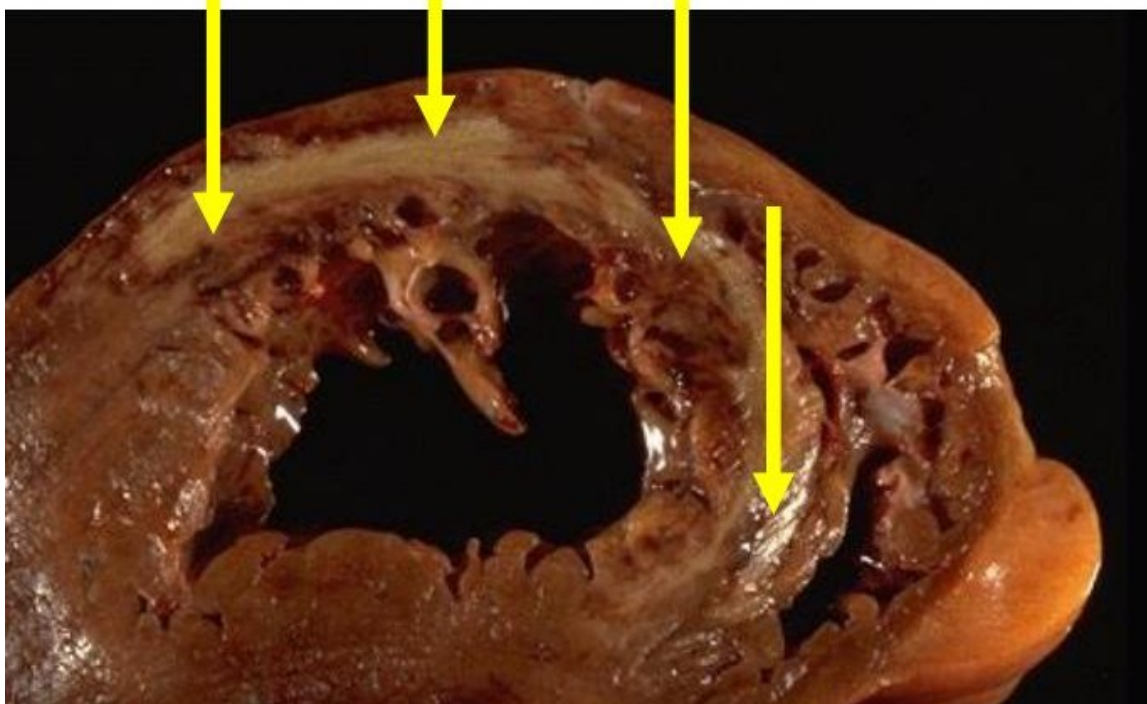


Figure L1.16 — Anteroseptal myocardial infarction: pale area of dead myocardium in the supply territory of the LAD.

ECG – ST-segment elevation anterior leads (T-wave inversion seen later); also a VE

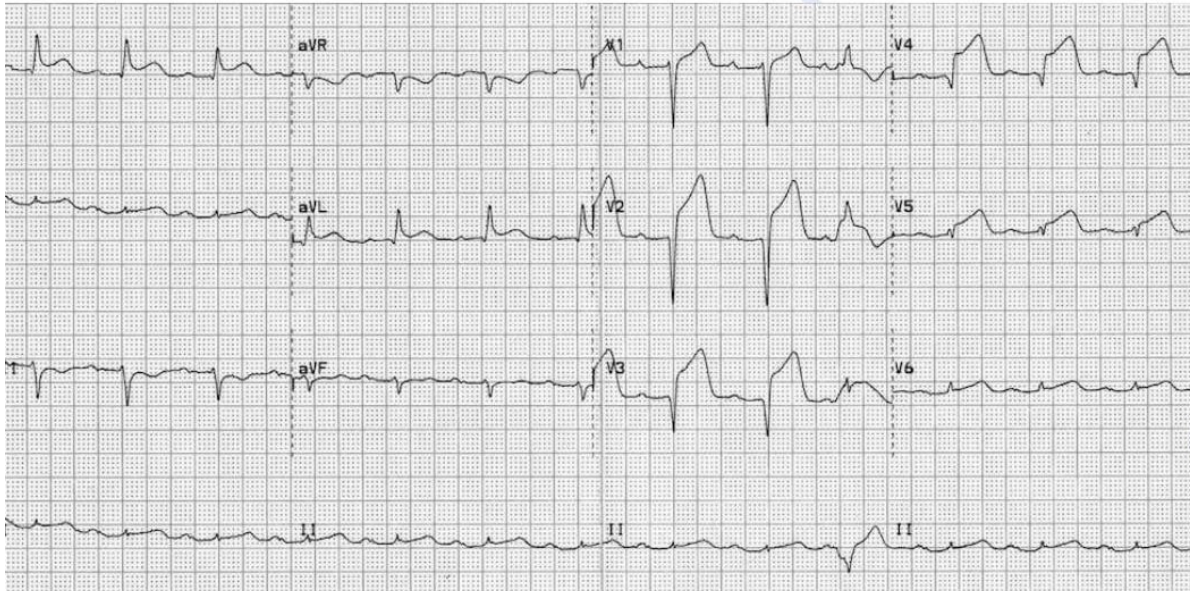


Figure L1.17 — ECG showing ST-segment elevation in anterior leads (anteroseptal STEMI) with a ventricular ectopic.

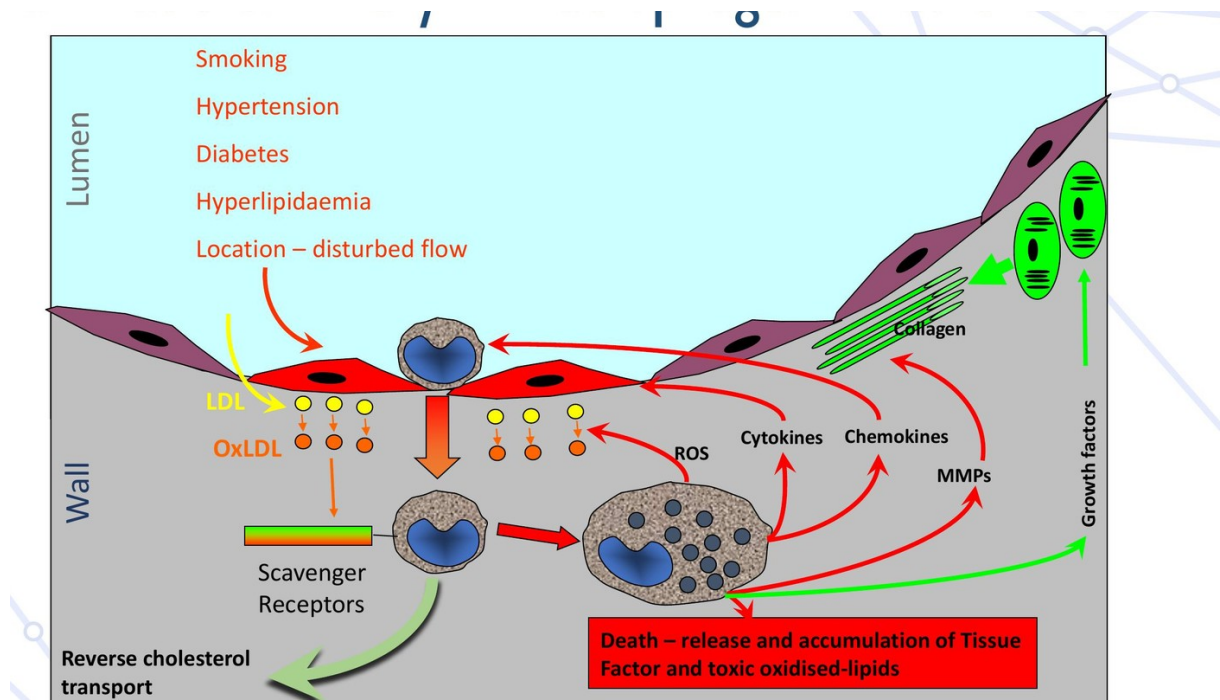


Figure L1.18 — Integrated diagram of plaque pathophysiology: risk factors converge on the endothelium, LDL enters and is oxidised, scavenger receptors drive foam-cell formation, foam cells secrete ROS/cytokines/chemokines/MMPs and growth factors, reverse cholesterol transport rescues some cholesterol, and foam cell death releases tissue factor and toxic oxylipids into the necrotic core.